Module 8: Planning — Using Math to Make Smarter Decisions

Learning Objectives

1. Learn how to use **mathematical predictions** to make better decisions about the future.
2. Understand **expected value** — a way of weighing options by considering both how likely outcomes are and how much they matter.
3. Apply mathematical planning to **real-life decisions** like budgeting, time management, and strategy games.

Introduction

You have a big project due in two weeks. You also have basketball practice four days a week, a birthday party this weekend, and three other classes with homework. How do you fit everything in without losing your mind?

This is a planning problem, and math can help you solve it. Not complicated math — just the practical kind. Estimating how long things take, figuring out what to prioritize, calculating whether you have enough time (or money, or energy) for everything you want to do.

Planning is where all your Math Detective skills come together. You're building a model (how will the next two weeks go?), making predictions (each section of the project will take about an hour), testing them against reality (the introduction actually took 90 minutes), and updating your plan accordingly. It's active inference applied to your schedule.

Key Concepts

1. Expected Value: Weighing Your Options

When you have multiple options, how do you choose? One powerful mathematical tool is **expected value** — a way of considering both how likely different outcomes are and how much they matter to you.

Simple example: A game offers you a choice. Option A: guaranteed \$5. Option B: flip a coin — heads you get \$12, tails you get \$0. Which should you pick?

The expected value of Option A is \$5 (it's certain). The expected value of Option B is (50% x \$12) + (50% x \$0) = \$6. Mathematically, Option B is better on average — but it comes with risk.

You use this kind of thinking all the time without realizing it. "Should I study an extra hour or sleep?" Your brain is (roughly) calculating: the expected benefit of studying (higher grade, with some probability) vs. the expected benefit of sleep (more energy tomorrow, almost certain). Neither answer is always right — it depends on the situation.

The point isn't to calculate expected values for every decision. It's to develop the habit of thinking: "What's likely to happen with each option, and how much does each outcome matter?"

2. Time Estimation (and Why You're Bad at It)

Here's a universal math fact about humans: we are terrible at estimating how long things take. Scientists call this the **planning fallacy** — the tendency to underestimate the time, cost, and difficulty of future tasks.

You think the essay will take an hour. It takes three. You think you can finish the project the night before. You can't. You think getting ready in the morning takes 10 minutes. It takes 25.

Why does this happen? Your brain simulates the **best-case scenario** (everything goes smoothly) and treats it as the prediction. But in reality, things go wrong. You get distracted. The printer runs out of ink. You realize you need a source you haven't found yet.

The mathematical fix: **multiply your initial estimate by 1.5 to 2**. If you think it'll take an hour, plan for 1.5 to 2 hours. This isn't pessimism — it's realism based on data. Track your time estimates vs. actual times for a week and you'll probably find that the 1.5x multiplier is pretty accurate.

3. Budgeting: Math Planning in Action

Budgeting your money or time is one of the purest forms of mathematical planning, and it uses every skill you've learned in this unit.

A budget is a mathematical model: `what I have - what I need to spend = what's left for wants`. Variables include income (allowance, gifts, earnings), needs (school supplies, transportation), and wants (games, snacks, entertainment).

The planning part comes from projecting into the future: If I save $10 per week, how many weeks until I can afford the $80 item I want? If I spend $5 on snacks every week, that's $260 per year — what else could I do with that money?

Budgeting also teaches you about **trade-offs** — the math of "if I choose this, I can't have that." Every dollar spent on one thing is a dollar not available for something else. Every hour spent on one activity is an hour not available for another. Understanding trade-offs mathematically helps you make decisions that align with what you actually care about, rather than just spending impulsively.

Try It!

The Two-Week Planner. Take your next two weeks and plan them mathematically. List everything you need to do (school, activities, obligations) with time estimates for each. Multiply each estimate by 1.5 for realism. Add up the total time and compare it to the total available hours (after subtracting sleep, meals, and travel). If the math doesn't work out, you need to adjust — cut something, delegate something, or accept that something won't get as much time as you'd like. This exercise often reveals why you feel stressed: there's literally more to do than hours to do it, and the math makes it visible.

Summary

Mathematical planning uses **expected value** to weigh options, **realistic time estimation** (multiply by 1.5!) to avoid the planning fallacy, and **budgeting** to manage resources and trade-offs. These skills combine everything you've learned as a Math Detective — systems thinking, pattern recognition, model building, testing, updating, and communication — and apply them to the decisions you make every day. You're not just learning math anymore. You're using math to navigate the world more effectively. Case closed, Detective.